Z+$b\bar{b}$ Study with Updated Samples

Samples, selection, preselection object diagnostics, and corrected result

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Samples

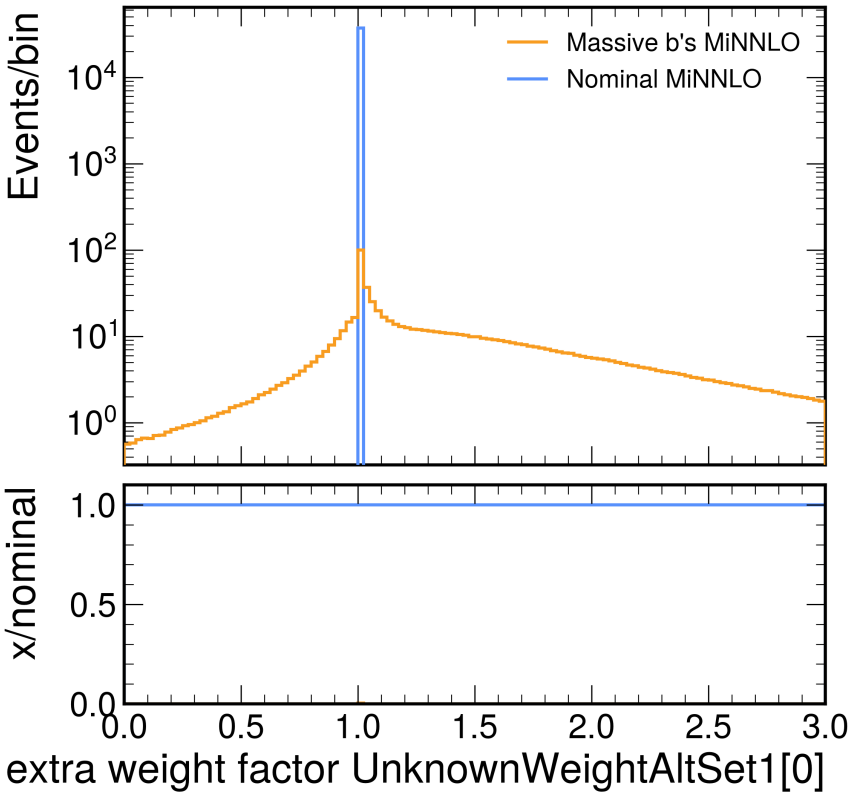
- Nominal sample (5FS, massless b): Zmumu MiNNLO.
- Alternate sample (4FS, massive b): Zbb MiNNLO.
- We now use the correct event weights from the LHE files: thanks to Marius, Kenneth for setting this up.
- Marco Zaro confirmed that in arXiv1803.04336 they used B hadrons to swap, as we do here.

Event Selection and Swap Definition

- Event selection:
  - prefsrLep_pt > 20
  - prefsrOtherLep_pt > 20
  - prefsrLep_absEta < 2.5
  - prefsrOtherLep_absEta < 2.5
  - abs(prefsrV_mass - 91.) < 15
- Event-level selected region for swap: $bottom_sel = (nBhad \geq 1)$. No requirements on the B hadrons themselves.
- B hadrons are built from final-state GenPart bottom-hadron ID logic, as before.
- Corrected sample construction: nominal unselected + massive selected.
- Slides below show both preselection LHE, b-tagged jet, and B hadron diagnostics.

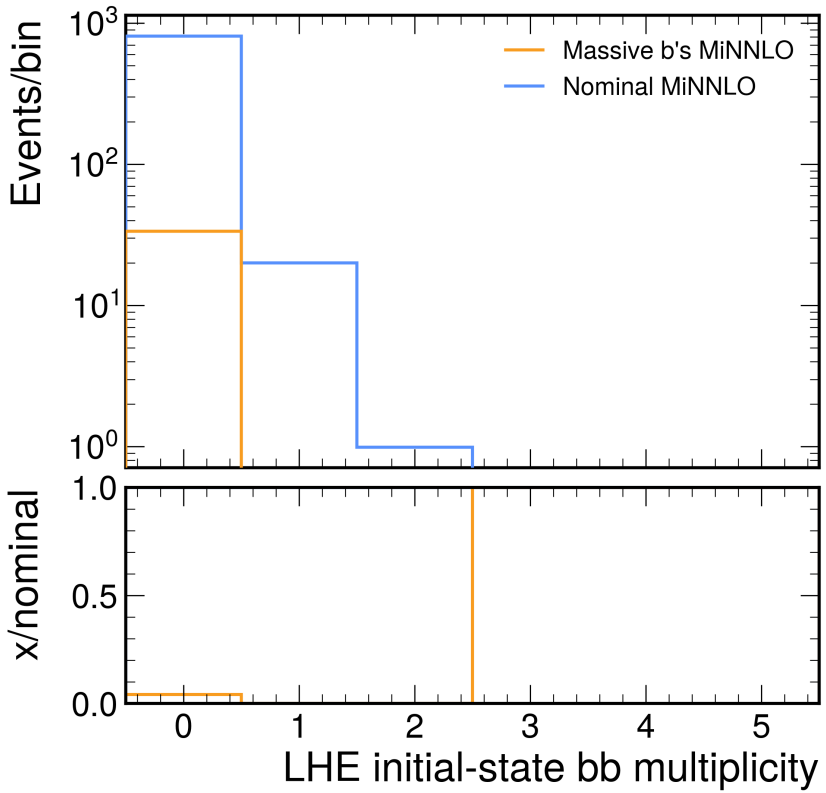
Event Weight Histogram

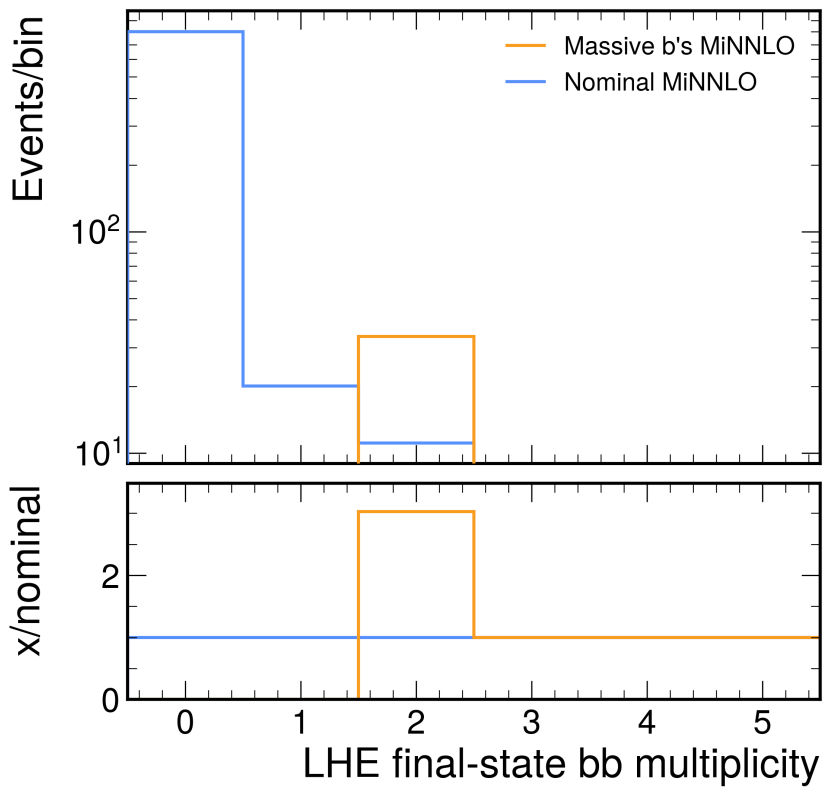
- Distribution of the updated LHE event weight used in this run.



Post-Cut LHE Multiplicity

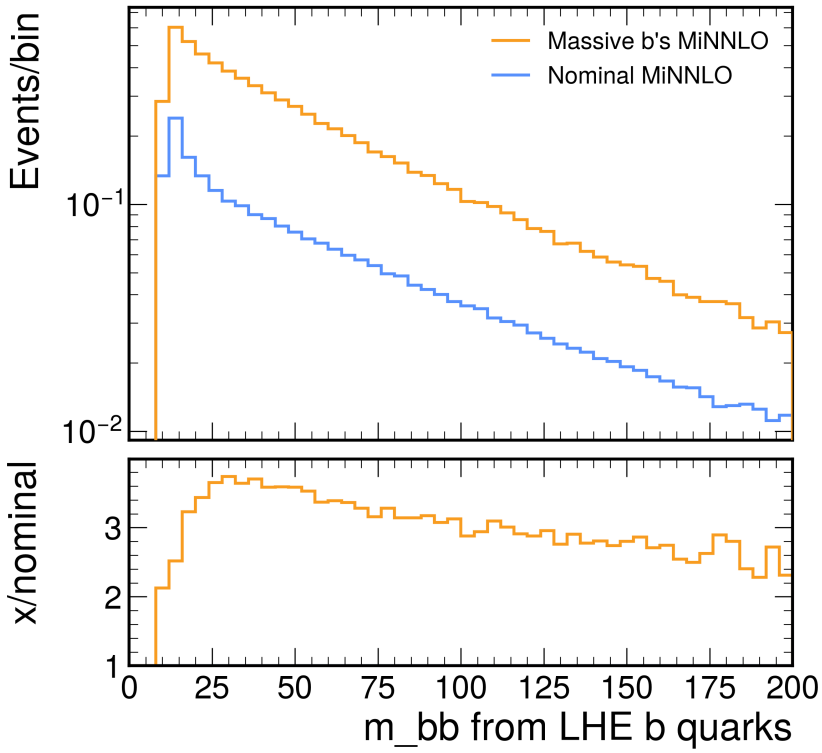
- After lepton/event selection cuts, inclusive in B-hadrons.

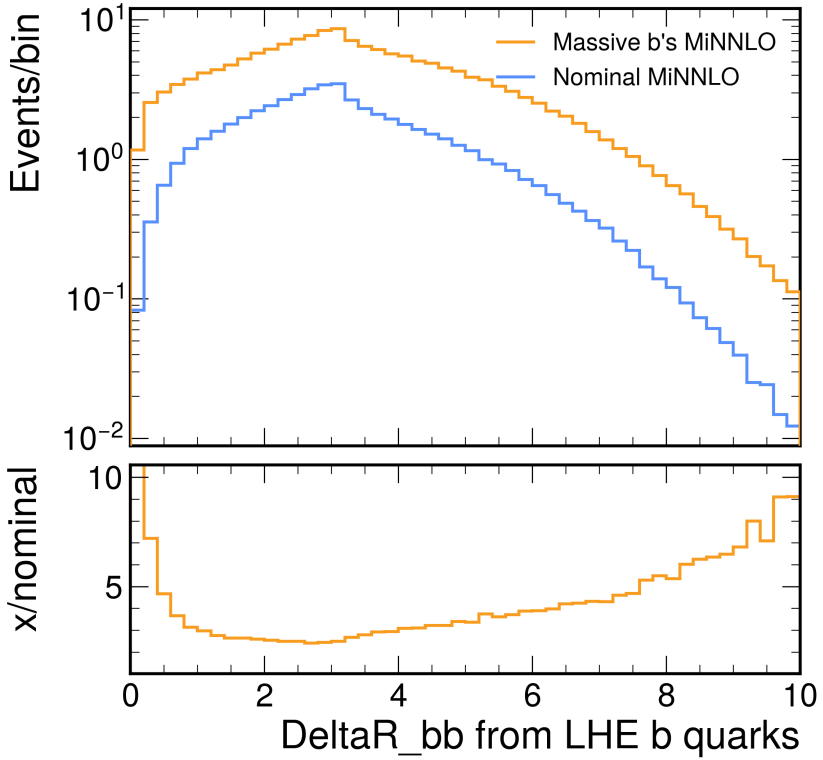




Post-Cut LHE Kinematics

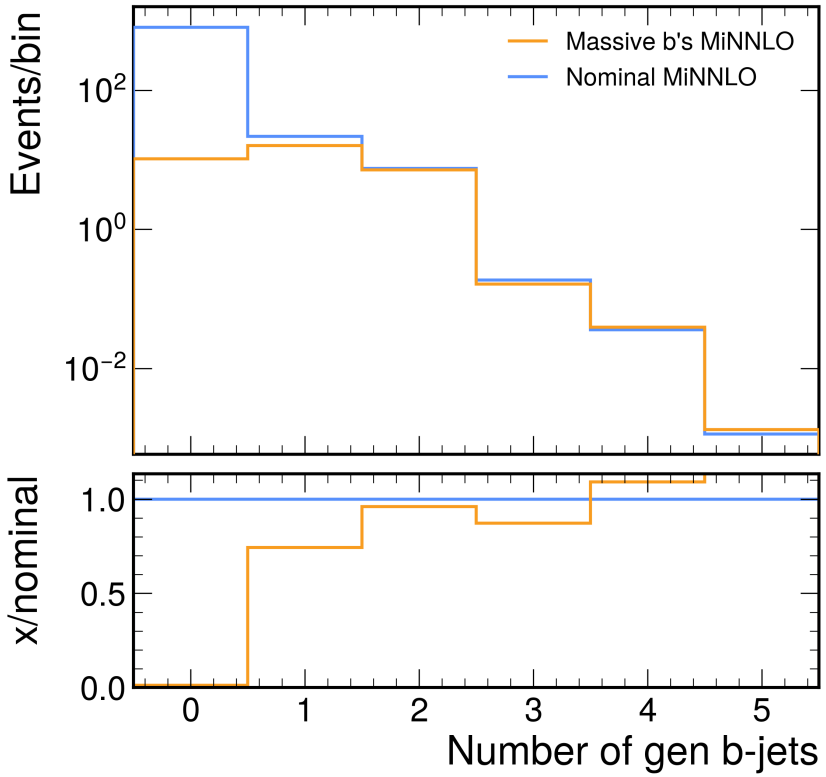
- After lepton/event selection cuts, inclusive in B-hadrons.

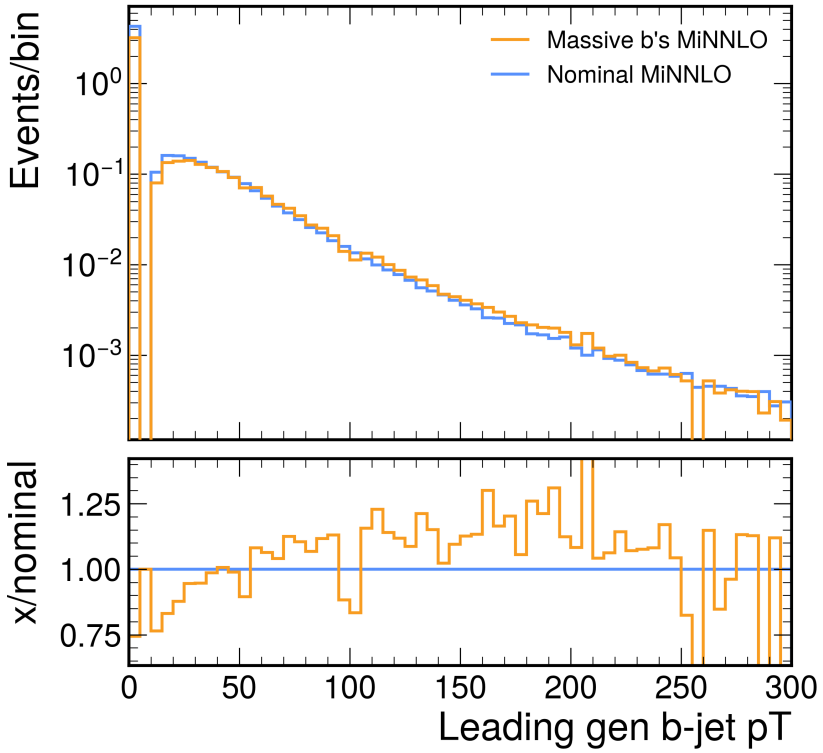




Selected-Region Gen Jet Overview

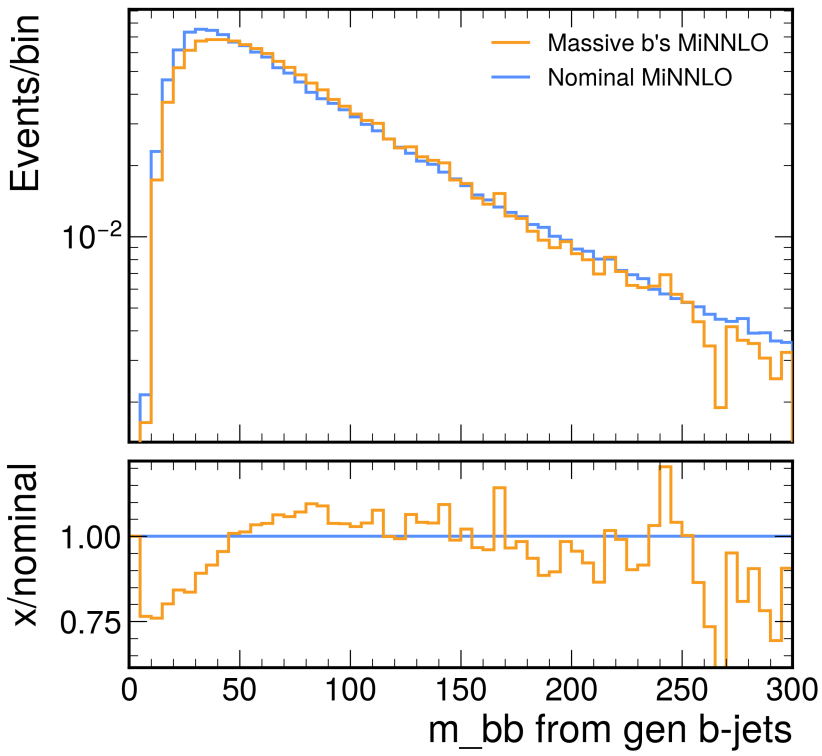
- After lepton/event selection cuts, inclusive in B-hadrons.

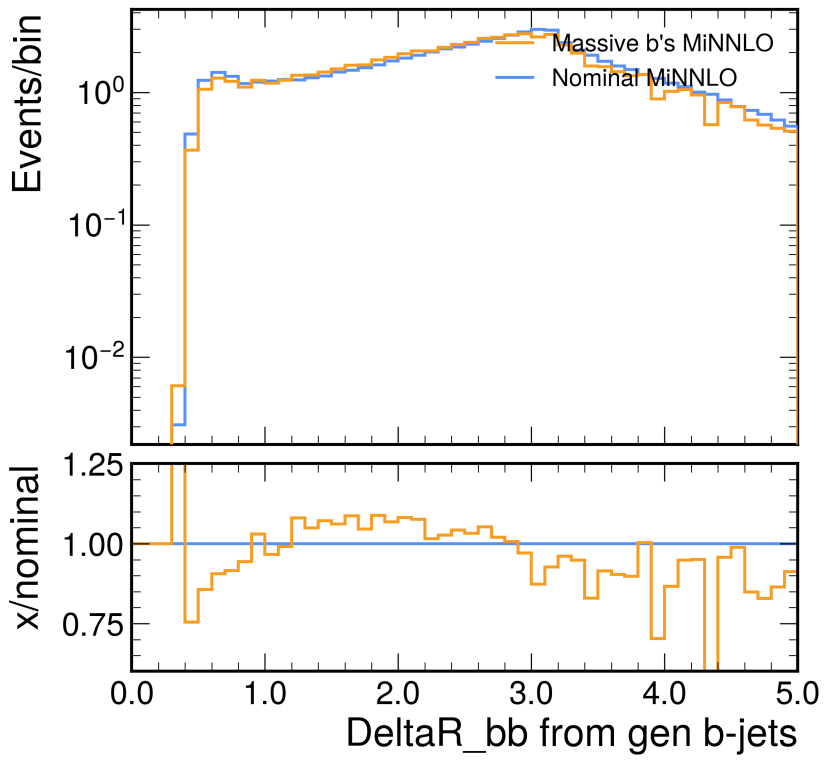




Selected-Region Gen Jet Pair Kinematics

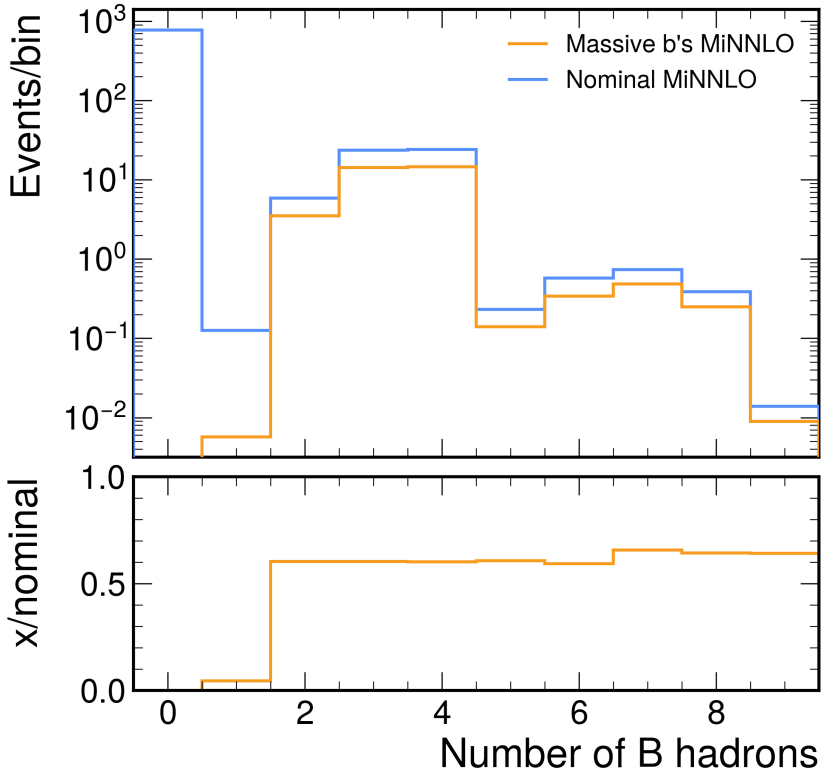
- Combination of jet-pair observables.

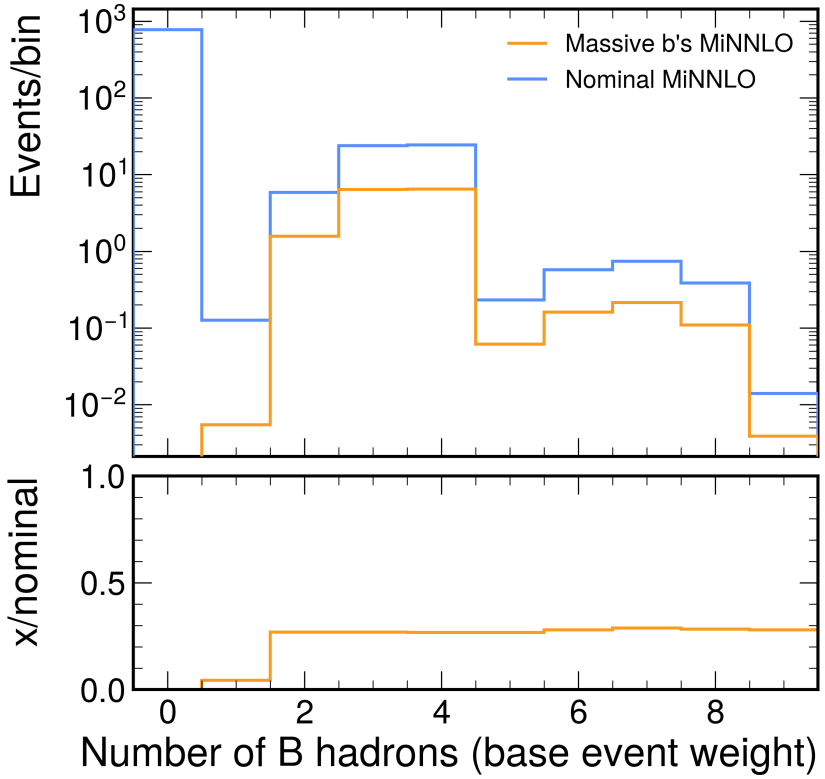




B-Hadron Multiplicity

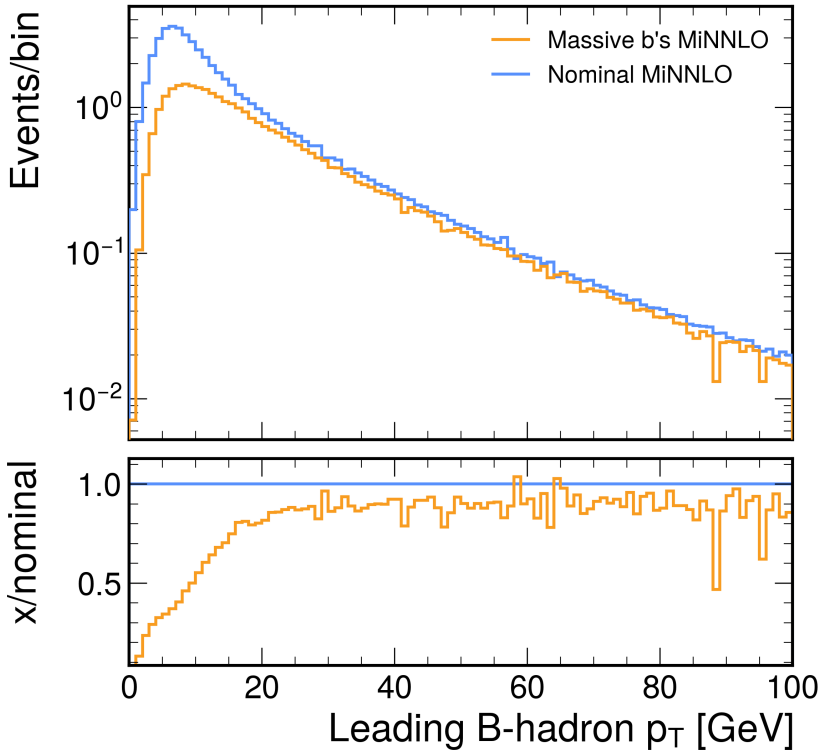
- Both panels use no B-hadron requirements, after basic event selection.
- Factor of ~2 increase in event yields from the updated weights.

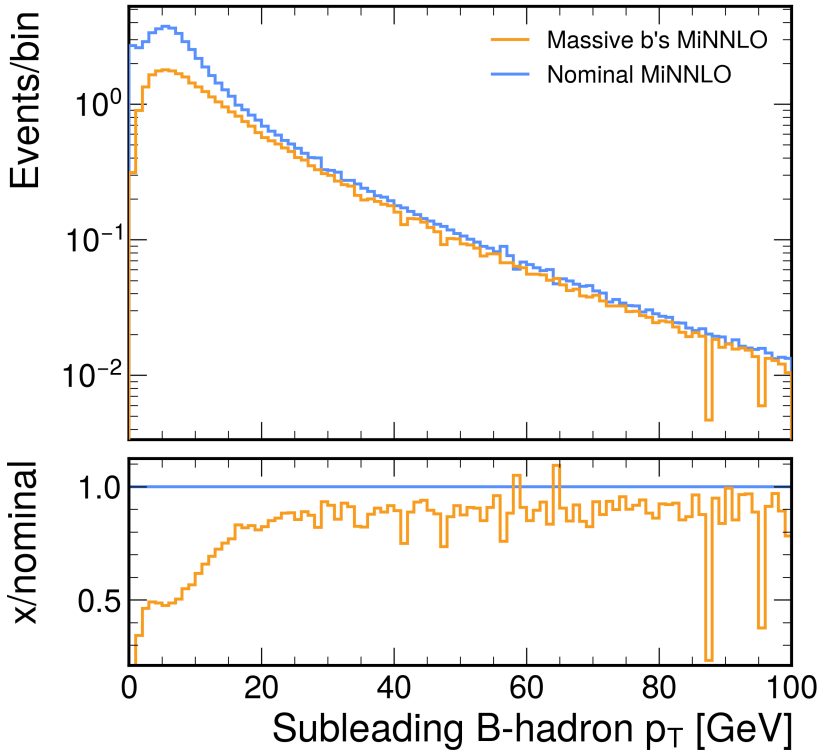




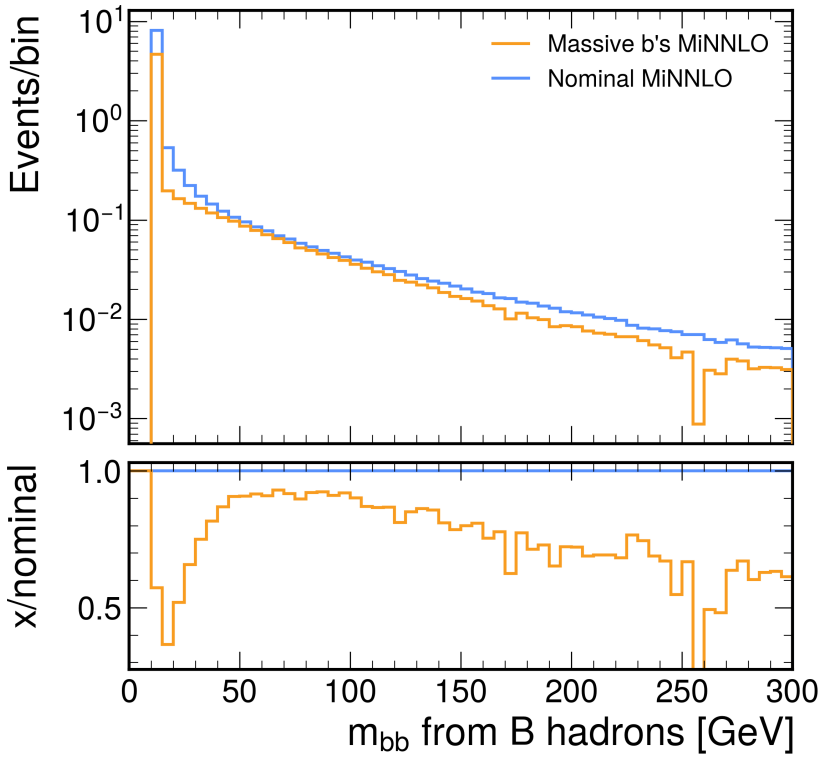
B-Hadron Single-Object Kinematics

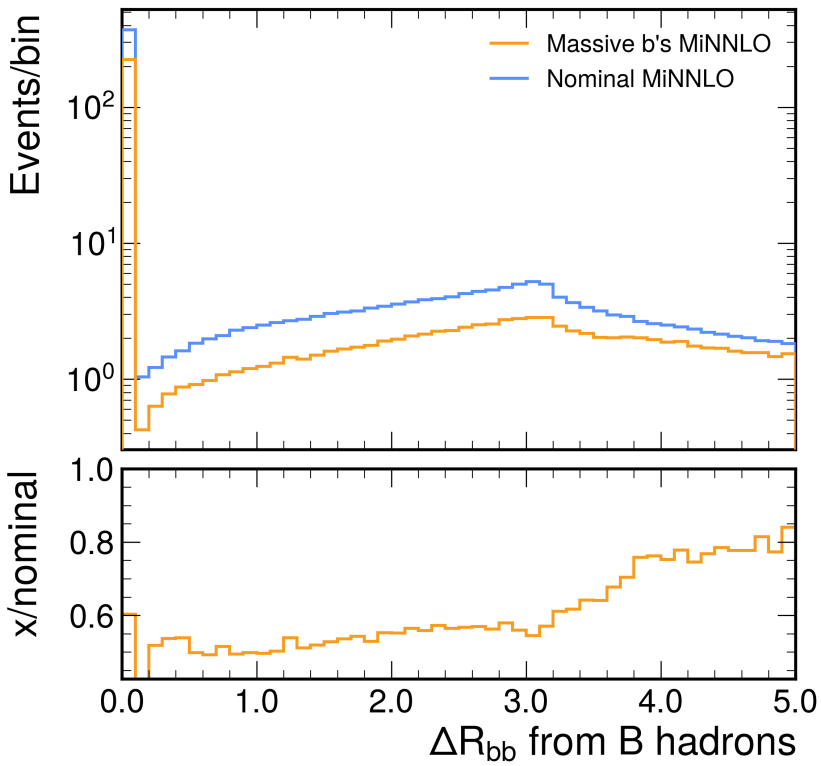
- Improved agreement between samples.
- Clear differences at low p_T. Physical?





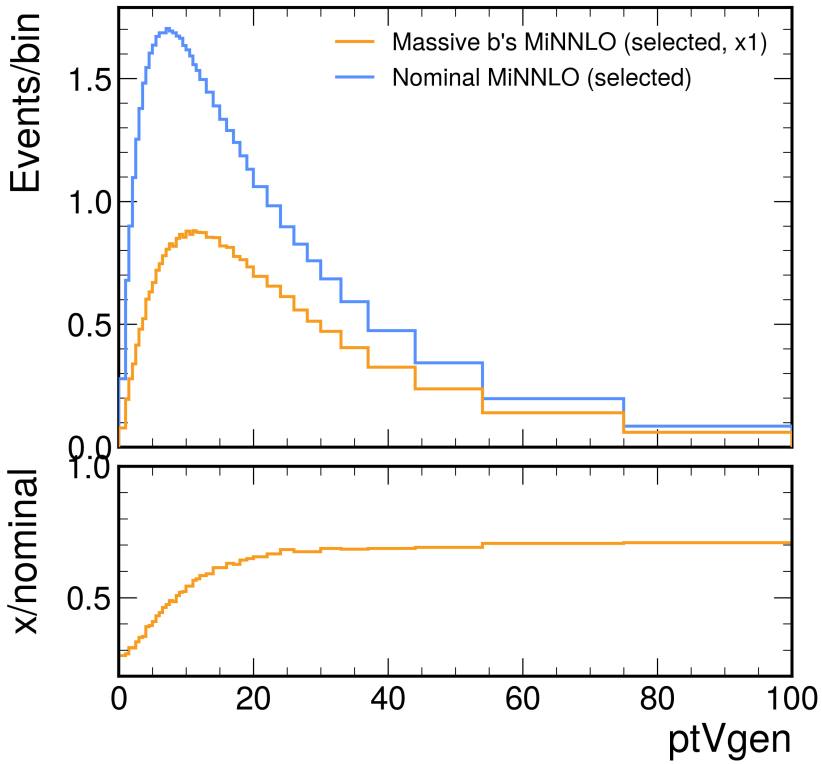
B-Hadron Pair Kinematics

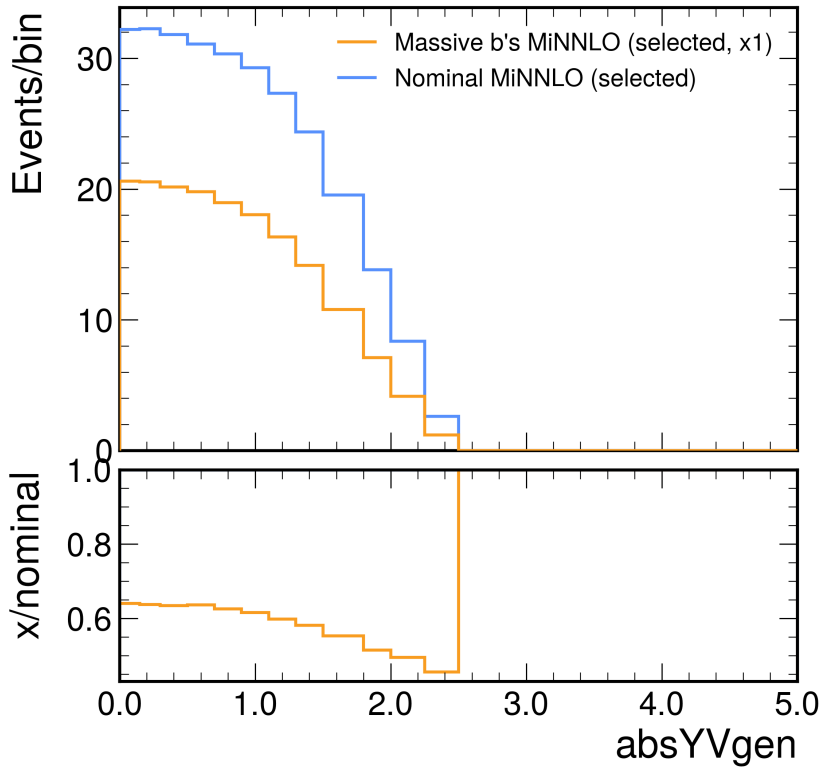


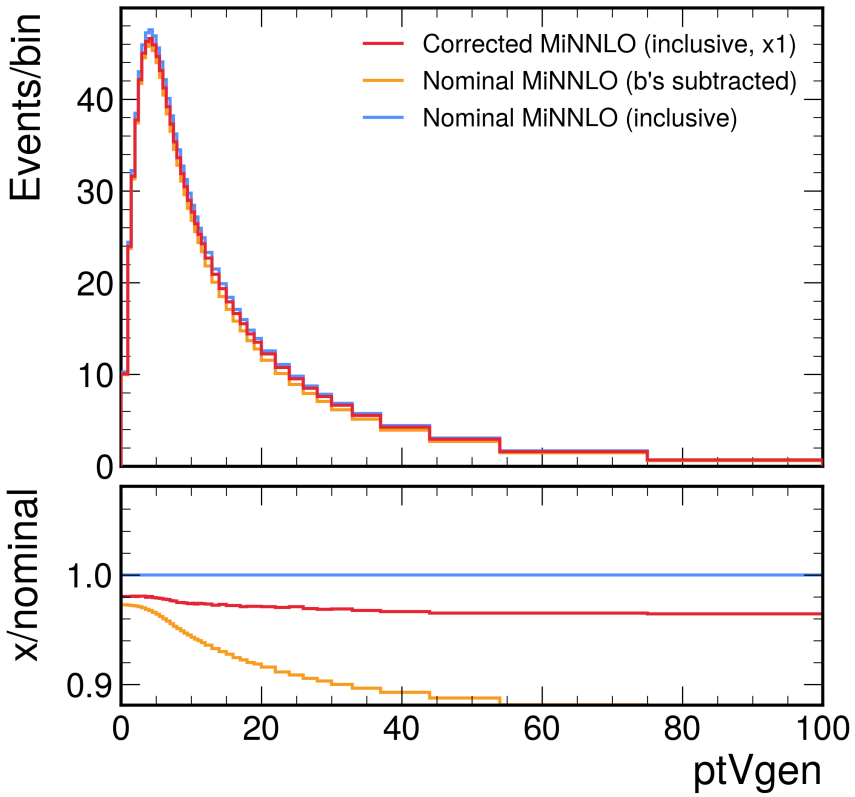


Corrected Sample: Selected Component

- Selected-component ($nBhad >= 1$) comparison in p_T^Z and $|y_Z|$.





Corrected Sample: Inclusive Z pT

- Inclusive corrected distribution from the current swap recipe.